

Level Up: Decoding Global Video Game Sales with SQL

Exploratory and Statistical Analysis of a Video Game Sales Database using MySQL



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Introduction

This project aims to analyze the behavior of video game sales at both global and regional levels, identifying relevant patterns by genre, platform, year, and publisher. To achieve this, SQL tools were used throughout the data exploration, cleaning, and analysis process, in order to extract key insights that help understand the dynamics of the market.

Database Structure

- **Database name:** video_game_sales
- **Main table:** vgsales
- **Key columns:** Name, Platform, Year, Genre, Publisher, Global_Sales, north_america_sales, europe_sales, japan_sales, other_sales
- **Number of rows:** 16,327
- **Time period covered:** 1980–2020 (except for the years 2018 and 2019)

rank	name	platform	year	genre	publisher	north_america_sales	europe_sales	japan_sales	other_sales	global_sales
1	Wii Sports	Wii	2006	Sports	Nintendo	41.49	29.02000	3.77000	8.46000	82.74000
2	Super Mario Bros.	NES	1985	Platform	Nintendo	29.08	3.58000	6.81000	0.77000	40.24000
3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.85	12.88000	3.79000	3.31000	35.82000
4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.75	11.01000	3.28000	2.96000	33.00000
5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	11.27	8.89000	10.22000	1.00000	31.37000
6	Tetris	GB	1989	Puzzle	Nintendo	23.2	2.26000	4.22000	0.58000	30.26000
7	New Super Mario Bros.	DS	2006	Platform	Nintendo	11.38	9.23000	6.50000	2.90000	30.01000
8	Wii Play	Wii	2006	Misc	Nintendo	14.03	9.20000	2.93000	2.85000	29.02000
9	New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	14.59	7.06000	4.70000	2.26000	28.62000
10	Duck Hunt	NES	1984	Shooter	Nintendo	26.93	0.63000	0.28000	0.47000	28.31000

Explanation of Key Fields

- **name:** Name of the video game.
- **platform:** Platform the game was released on (e.g., PlayStation, X360).
- **year:** Year of release.
- **genre:** Genre of the game (e.g., Action, Adventure).
- **publisher:** Company that published the game (e.g., Activision, Nintendo).
- **north_america_sales, europe_sales, japan_sales, other_sales, global_sales:** Sales (in millions of units) in North America, Europe, Japan, other countries, and globally, respectively.

Note: This table contains data on over 16,000 video games.

Data Quality Analysis

Objective: Identify records with missing or inconsistent data.

Query:

```
SELECT * FROM vgsales WHERE `rank` IS NULL OR `name` IS NULL OR platform IS NULL OR genre IS NULL OR publisher IS NULL OR north_america_sales IS NULL OR europe_sales IS NULL OR japan_sales IS NULL OR other_sales IS NULL OR global_sales IS NULL;
```

Result:

	rank	name	platform	year	genre	publisher	north_america_sales	europe_sales	japan_sales	other_sales	global_sales
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Query:

```
SELECT `rank`, COUNT(`rank`) AS frequency FROM vgsales GROUP BY `rank` HAVING frequency > 1;
```

Result:

	rank	frequency
--	------	-----------

Analysis:

We observe that there are no NULL values or duplicate rows. Based on these and other queries, we conclude that the data is consistent.

Descriptive Statistics of Sales by Region and Globally

Analysis of sales statistics by region (North America, Europe, Japan, and Others), along with a global metric summarizing worldwide sales. Metrics include maximum, minimum, average sales, standard deviation, and coefficient of variation.

region	max_sale	min_sale	average_sale	standard_desviation	coefficient_of_variation
Global	82.74	0.01	0.5402	1.5657	289.82
North_America	41.49	0	0.2654	0.8216	309.54
Europe	29.02	0	0.1476	0.5088	344.79
Others	10.57	0	0.0483	0.1899	392.92
Japan	10.22	0	0.0787	0.3115	396.06

Analysis:

We can observe that the region with the highest individual sale is North America, with 41.49 million units sold, followed by Europe with 29.02 million, then Other Countries and Japan with 10.57 and 10.22 million respectively. An interesting insight comes from the mean values, which are significantly lower than the maximums. This suggests that most sales figures are low, with only a few games reaching very high numbers. We also note that the coefficients of variation are quite high, ranging from 289.82% to 396.06%, indicating a large spread in the data.

Top 10 Best-Selling Video Games Worldwide

Objective:

Identify the top-selling video games globally, considering as separate entries those with the same name but released on different platforms.

name	platform	year	genre	publisher	global_sales
Wii Sports	Wii	2006	Sports	Nintendo	82.74
Super Mario Bros.	NES	1985	Platform	Nintendo	40.24
Mario Kart Wii	Wii	2008	Racing	Nintendo	35.82
Wii Sports Resort	Wii	2009	Sports	Nintendo	33.00
Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	31.37
Tetris	GB	1989	Puzzle	Nintendo	30.26
New Super Mario Bros.	DS	2006	Platform	Nintendo	30.01
Wii Play	Wii	2006	Misc	Nintendo	29.02
New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	28.62
Duck Hunt	NES	1984	Shooter	Nintendo	28.31

Analysis: It is evident that Nintendo overwhelmingly dominates the list of historically best-selling games, being the only publisher represented in this top 10.

Additionally, Mario Bros appears the most frequently on the list, with a total of three entries. This highlights the enduring popularity of this classic game that continues to captivate players around the world.

Top-Selling Video Game Genres Globally

SQL Query:

```
SELECT genre,  
ROUND(SUM(global_sales), 2) AS  
total_global_sales FROM vgsales  
GROUP BY genre ORDER BY  
total_global_sales DESC;
```

Analysis:

The most popular genres are Action, Sports, and Shooter, respectively. These are the only ones to surpass one billion units sold globally.

genre	total_global_sales
Action	1722.88
Sports	1309.24
Shooter	1026.20
Role-Playing	923.84
Platform	829.15
Misc	797.62
Racing	726.77
Fighting	444.05
Simulation	390.16
Puzzle	242.22
Adventure	234.80
Strategy	173.43

Top-Selling Video Game Platforms Globally

platform	total_global_sales
PS2	1233.46
X360	969.61
PS3	949.35
Wii	909.81
DS	818.96
PS	727.39
GBA	313.56
PSP	291.71
PS4	278.10
PC	255.05

Analysis:

The best-selling platform worldwide is the PlayStation 2 (PS2), being the only one to surpass one billion units sold, with a total of 1,233.46 million.

Next in line are the X360, PS3, and Wii, with sales of 969.61, 949.35, and 909.81 million, respectively.

We observe a clear “split” in the table starting from the PS and GBA platforms, where the PS (727.39 million) more than doubles the GBA’s 313.56 million in sales. This highlights the significant advantage held by platforms such as the PS2, X360, PS3, Wii, DS, and PS over the rest.

Top 10 Publishers with the Highest Global Sales

publisher	total_global_sales
Nintendo	1784.43
Electronic Arts	1093.39
Activision	721.41
Sony Computer Entertainment	607.28
Ubisoft	473.54
Take-Two Interactive	399.30
THQ	340.44
Konami Digital Entertainment	278.56
Sega	270.70
Namco Bandai Games	253.65

Objective: Identify the most successful publishers in terms of global sales to understand which companies have historically dominated the video game market.

Analysis:

Nintendo is the top-selling publisher over the years, with a total of 1,784.43 million units sold.

Electronic Arts follows in second place with 1,093.39 million units. These two companies are the only ones to surpass the one-billion mark.

Further down the ranking, notable names include **Activision, Sony Computer Entertainment, and Ubisoft**, with 721.41 million and 473.54 million units, respectively—solidifying their historical impact on the industry.

Golden Years of Global Video Game Sales

Objective:

Identify the years with the highest volume of global video game sales to detect historical trends, peak periods in the industry, and their possible relationship with key releases or market shifts.

Analysis:

We can deduce that the golden era of the video game industry occurred between 2004 and 2011. The peak year was 2008, with total sales reaching 678.9 million units, closely followed by 2009 with 667.3 million units.

year	total_global_sales
2008	678.90
2009	667.30
2007	611.13
2010	600.45
2006	521.04
2011	515.99
2005	459.94
2004	419.31
2002	395.52
2013	368.11

Leading Platform in Sales by Year

Every year, one platform dominated global sales. This analysis aims to identify trends and market behavior regarding platforms:

Analysis:

- **1980s:** The classic Atari 2600 and NES were the undisputed market leaders.
- **1990s:** Nintendo initially dominated with the SNES, followed by PlayStation taking the top spot starting in 1995 and consolidating its position until the end of the decade.
- **2000s:** PlayStation 2 reigned as the absolute king, later followed by the rise of Nintendo Wii and Xbox 360.
- **2010s and beyond:** PlayStation 3 and later PlayStation 4 remained the absolute leaders until 2017, and then DS took over in 2020.
- **Since 2017,** general sales have significantly dropped compared to previous years, mainly due to shifts in the industry, such as the rise of mobile and digital games.

# year	total global sales	platform
1980	1.138.000	2600
1981	3.577.000	2600
1982	2.886.000	2600
1983	1.096.000	NES
1984	5.009.000	NES
1985	5.344.000	NES
1986	3.641.000	NES
1987	1.976.000	NES
1988	4.501.000	NES
1989	6.498.000	GB
1990	2.616.000	SNES
1991	1.621.000	SNES
1992	3.298.000	SNES
1993	4.001.000	SNES
1994	3.508.000	SNES
1995	3.592.000	PS
1996	9.468.000	PS
1997	13.608.000	PS
1998	16.958.000	PS
1999	14.457.000	PS
2000	9.628.000	PS
2001	16.643.000	PS2
2002	20.540.000	PS2
2003	18.429.000	PS2
2004	21.178.000	PS2
2005	16.065.000	PS2
2006	13.791.000	Wii
2007	15.497.000	Wii
2008	17.416.000	Wii
2009	21.044.000	Wii
2010	17.105.000	X360
2011	15.937.000	PS3
2012	10.949.000	PS3
2013	11.739.000	PS3
2014	9.876.000	PS4
2015	11.530.000	PS4
2016	3.925.000	PS4
2017	0.03000	PS4
2020	0.29000	DS

year	publisher	total_sales
1980	Atari	8.36
1981	Activision	8.50
1982	Atari	19.43
1983	Nintendo	10.96
1984	Nintendo	45.56
1985	Nintendo	49.95
1986	Nintendo	16.18
1987	Nintendo	11.95
1988	Nintendo	36.44
1989	Nintendo	63.88
1990	Nintendo	35.49
1991	Nintendo	15.97
1992	Nintendo	38.11
1993	Nintendo	20.04
1994	Nintendo	24.99
1995	Sony Computer Entertainment	18.45
1996	Nintendo	73.70
1997	Sony Computer Entertainment	43.90
1998	Nintendo	48.41
1999	Nintendo	65.33
2000	Nintendo	34.05
2001	Nintendo	45.37
2002	Electronic Arts	73.01
2003	Electronic Arts	69.83
2004	Electronic Arts	67.33
2005	Nintendo	127.47
2006	Nintendo	205.61
2007	Nintendo	104.18
2008	Nintendo	91.22
2009	Nintendo	128.89
2010	Electronic Arts	81.38
2011	Electronic Arts	72.25
2012	Nintendo	56.47
2013	Electronic Arts	52.92
2014	Nintendo	48.65
2015	Electronic Arts	45.42
2016	Electronic Arts	12.25
2017	Sega	0.04
2020	Ubisoft	0.29

Most Successful Publisher by Year

This analysis aims to understand how market and industry changes have favored certain publishers over different periods, which can be useful for anticipating trends and understanding the industry's evolution.

Analysis:

1980s and 1990s: Nintendo dominated almost without competition, except for a few years at the end of the 90s when Sony took the top spot.

2000-2016: Electronic Arts and Nintendo alternated the first place, showing how competitive the race was between these two companies.

2017-2020: Sega and Ubisoft emerged, breaking the dominance of Nintendo and Electronic Arts.

Best-Selling Game by Year (1980–2020)

Analysis:

1980-1990: Atari and arcade-style games dominated (games like Asteroids, Pac-Man, Duck Hunt).

1990-2000: During these years, Nintendo dominated with Mario, The Legend of Zelda, and Pokémon.

2000-2010: The rise of open-world franchises and shooters: GTA, Need for Speed, and Call of Duty.

2010-2015: Call of Duty led year after year.

2016-2020: Leadership became more scattered, with much lower sales, showing the shift toward mobile games and digital downloads.

#year	name	total_sales
1980	Asteroids	4.31
1981	Pitfall!	4.50
1982	Pac-Man	7.81
1983	Baseball	3.20
1984	Duck Hunt	28.31
1985	Super Mario Bros.	40.24
1986	The Legend of Zelda	6.51
1987	Zelda II: The Adventure of Link	4.38
1988	Super Mario Bros. 3	17.28
1989	Tetris	30.26
1990	Super Mario World	20.61
1991	The Legend of Zelda: A Link to the Past	4.61
1992	Super Mario Land 2: 6 Golden Coins	11.18
1993	Super Mario All-Stars	10.55
1994	Donkey Kong Country	9.30
1995	Donkey Kong Country 2: Diddy's Kong Quest	5.15
1996	Pokemon Red/Pokemon Blue	31.37
1997	Gran Turismo	10.95
1998	Pokémon Yellow: Special Pikachu Edition	14.64
1999	Pokemon Gold/Pokemon Silver	23.10
2000	Pokémon Crystal Version	6.39
2001	Gran Turismo 3: A-Spec	14.98
2002	Grand Theft Auto: Vice City	16.15
2003	Need for Speed Underground	10.46
2004	Grand Theft Auto: San Andreas	20.81
2005	Nintendogs	24.76
2006	Wii Sports	82.74
2007	Wii Fit	22.72
2008	Mario Kart Wii	35.82
2009	Wii Sports Resort	33.00
2010	Call of Duty: Black Ops	29.40
2011	Call of Duty: Modern Warfare 3	30.83
2012	Call of Duty: Black Ops II	29.72
2013	Grand Theft Auto V	37.78
2014	Call of Duty: Advanced Warfare	21.90
2015	Call of Duty: Black Ops 3	25.32
2016	FIFA 17	6.91
2017	Phantasy Star Online 2 Episode 4: Deluxe Package	0.04
2020	Imagine: Makeup Artist	0.29

Top 10 Games with Highest Relative Sales Frequency

name	sales_by_game	relative_frequency_percent
Wii Sports	82.74	0.94
Grand Theft Auto V	55.92	0.63
Super Mario Bros.	45.31	0.51
Mario Kart Wii	35.82	0.41
Tetris	35.84	0.41
Wii Sports Resort	33.00	0.37
Pokemon Red/Pokemon Blue	31.37	0.36
Call of Duty: Modern Warfare 3	30.83	0.35
New Super Mario Bros.	30.01	0.34
Call of Duty: Black Ops II	29.72	0.34

Objective: Show which games concentrated the largest share of total sales historically.

Analysis:

- "Wii Sports" and "GTA V" are phenomena that broke all records.
- Nintendo dominates the list with multiple titles.
- It is evident how fragmented the market is: the top-selling game represents less than 1% of total game sales. Keep in mind that there are over 16,000 games on this list.

Relative Frequency of Global Sales by Genre

Analysis:

The most popular game genres are Action, Sports, and Shooter, representing 19.53%, 14.84%, and 11.63% of the market, respectively, with sales above one billion units.

Next are the Role-Playing games, representing 10.47%, Platform games at 9.4%, and Miscellaneous at 9.04%, with sales of 923.84 million for Role-Playing, 829.15 million for Platform, and 796.77 million for Miscellaneous. Genres like Puzzle, Adventure, and Strategy represent a much smaller part of the market, each with less than 3% market share.

genre	sales_by_genre	relative_frequency_percent
Action	1722.88	19.53
Sports	1309.24	14.84
Shooter	1026.20	11.63
Role-Playing	923.84	10.47
Platform	829.15	9.40
Misc	797.62	9.04
Racing	726.77	8.24
Fighting	444.05	5.03
Simulation	390.16	4.42
Puzzle	242.22	2.75
Adventure	234.80	2.66
Strategy	173.43	1.97

Relative Frequency of Global Sales by Platform

Analysis:

The most popular platforms in terms of global sales are traditional consoles such as PS2, X360, and PS3, with PS2 leading the market, holding a market share of 13.98%.

On the other hand, Wii and DS continue to be key platforms, with Wii representing 10.31% and DS 9.28% of the global market.

More modern and recent platforms like PS4 remain highly relevant, but their market share is smaller compared to the previously mentioned consoles (PS2 and X360).

In 10th place is PC, which has a relatively modest market share compared to consoles, representing only 2.89% of global sales.

platform	sales_by_platform	relative_frequency_percent
PS2	1233.46	13.98
X360	969.61	10.99
PS3	949.35	10.76
Wii	909.81	10.31
DS	818.96	9.28
PS	727.39	8.25
GBA	313.56	3.55
PSP	291.71	3.31
PS4	278.10	3.15
PC	255.05	2.89

region	sales_by_region	relative_frequency_percent
north_america	4333.43	49.13
europa	2409.12	27.31
japan	1284.30	14.56
others	789.01	8.95

Relative Frequency of Global Sales by Region

Analysis:

North America: Represents 49.13% of global sales, solidifying its position as the dominant region. It leads by a significant margin over the others.

Europe: Follows in the top with 27.31% of the market, positioning itself as the second most important region.

Japan: With 14.56% of the total global sales, it represents a considerable portion. Although it does not lead, it remains a key region.

Other countries: Encompasses all other regions or countries. With a share of 8.95%, it indicates that other areas of the world have a smaller impact on the global video game market.

Publishers with the Most Games in the Top 50 Global Sales

publisher	publisher_frequency
Nintendo	32
Activision	8
Take-Two Interactive	6
Microsoft Game Studios	2
Sony Computer Entertainment	2

Analysis:

- **Nintendo** leads this ranking with 32 games in the Top 50, featuring franchises like Super Mario, Zelda, and Pokémon.
- **Activision** is second, with 8 games in the Top 50, leading with the Call of Duty series.
- **Take-Two Interactive** has 6 games in the ranking, with titles like Grand Theft Auto standing out.
- **Microsoft Game Studios and Sony Computer Entertainment** each have 2 games, with franchises like Halo and Gran Turismo.
- It's worth noting that **Electronic Arts** (the second-largest publisher by global sales) has no games in the Top 50. This could be attributed to their business model, where most of their sales come from many moderately successful games rather than a few massive hits.

Top 10 Best-Selling Games by Region

North America			Europe		
name	platform	north_america_sales	name	platform	europe_sales
Wii Sports	Wii	41.49	Wii Sports	Wii	29.02
Super Mario Bros.	NES	29.08	Mario Kart Wii	Wii	12.88
Duck Hunt	NES	26.93	Wii Sports Resort	Wii	11.01
Tetris	GB	23.2	Nintendogs	DS	11.00
Mario Kart Wii	Wii	15.85	Grand Theft Auto V	PS3	9.27
Wii Sports Resort	Wii	15.75	Brain Age: Train Your Brain in Minutes a Day	DS	9.26
Kinect Adventures!	X360	14.97	New Super Mario Bros.	DS	9.23
New Super Mario Bros. Wii	Wii	14.59	Wii Play	Wii	9.20
Wii Play	Wii	14.03	Pokemon Red/Pokemon Blue	GB	8.89
Super Mario World	SNES	12.78	Wii Fit Plus	Wii	8.59

Analysis:

In **North America**, Nintendo dominates: Wii Sports, Super Mario Bros., and Duck Hunt lead the ranking, all with sales over 25 million.

In **Europe**, Wii Sports also leads, and we also see the rise of Nintendogs and Grand Theft Auto V in this continent.

Top 10 Best-Selling Games by Region

Japan			Others		
name	platform	japan_sales	name	platform	other_sales
Pokemon Red/Pokemon Blue	GB	10.22	Grand Theft Auto: San Andreas	PS2	10.57
Pokemon Gold/Pokemon Silver	GB	7.20	Wii Sports	Wii	8.46
Super Mario Bros.	NES	6.81	Gran Turismo 4	PS2	7.53
New Super Mario Bros.	DS	6.50	Grand Theft Auto V	PS3	4.14
Pokemon Diamond/Pokemon Pearl	DS	6.04	Mario Kart Wii	Wii	3.31
Pokemon Black/Pokemon White	DS	5.65	Wii Sports Resort	Wii	2.96
Pokemon Ruby/Pokemon Sapphire	GBA	5.38	Pro Evolution Soccer 2008	PS2	2.93
Animal Crossing: Wild World	DS	5.33	New Super Mario Bros.	DS	2.90
Brain Age 2: More Training in Minutes a Day	DS	5.32	Wii Play	Wii	2.85
Monster Hunter Freedom 3	PSP	4.87	Nintendogs	DS	2.75

Analysis:

In **Japan**, they love Pokémon: 6 of the 10 best-selling games are from that series. The ranking also features classics like Super Mario Bros and New Super Mario Bros.

In the **Other Countries group**, Grand Theft Auto: San Andreas stands out as the favorite, followed by Wii Sports and PlayStation 2 hits like Gran Turismo 4.

Analysis of Best-Selling Genres by Region

Norte América

genre	total_na
Action	861.8
Sports	670.09
Shooter	575.16
Platform	445.99
Misc	402.48
Racing	356.93
Role-Playing	326.5
Fighting	220.74
Simulation	181.78
Puzzle	122.01
Adventure	102.06
Strategy	67.89

Europa

genre	total_eu
Action	516.48
Sports	371.34
Shooter	310.45
Racing	236.32
Misc	213.82
Platform	200.67
Role-Playing	187.58
Simulation	113.20
Fighting	100.00
Adventure	63.79
Puzzle	50.53
Strategy	44.94

Japón

genre	total_jp
Role-Playing	350.29
Action	158.66
Sports	134.76
Platform	130.65
Misc	106.67
Fighting	87.15
Simulation	63.54
Puzzle	56.68
Racing	56.61
Adventure	52.01
Strategy	49.10
Shooter	38.18

Otros

genre	total_others
Action	184.92
Sports	132.65
Shooter	101.90
Racing	76.68
Misc	74.02
Role-Playing	59.38
Platform	51.51
Fighting	36.19
Simulation	31.36
Adventure	16.70
Puzzle	12.47
Strategy	11.23

Analysis:

In **North America**, the Action genre leads by a large margin (861.8M), followed by Sports (670.09M) and Shooter (575.16M). Fast-paced, competitive genres clearly dominate.

In **Europe**, the trend is very similar: Action leads with 516.48M, but Racing holds a more significant position than in other regions, showing the continent's preference for racing and driving games.

In **Japan**, things change completely: Role-Playing (RPG) is the favorite genre, leading by a considerable margin with 350.29M in sales. This highlights the strong tradition of narrative-driven and long adventure games in this market.

In **Other countries**, Action dominates again with 184.92M, although overall sales are considerably lower. Sports and Shooter complete the top three with 132.65M and 101.90M, respectively.

name	total_na	total_eu	total_jp	total_others	total_global	biggest_sale
Zyuden Sentai Kyoryuger: Game de Gaburincho!!	0	0.00	0.05	0.00	0.05	japan
Zwei!!	0	0.00	0.02	0.00	0.02	japan
Zumba Fitness: World Party	0.31	0.16	0.00	0.05	0.50	north_america
Zumba Fitness Rush	0	0.16	0.00	0.02	0.19	europa
Zumba Fitness Core	0	0.13	0.00	0.02	0.14	europa
Zumba Fitness 2	1.54	1.07	0.00	0.28	2.88	north_america
Zumba Fitness	5.52	3.33	0.00	0.94	9.79	north_america
Zubo	0.09	0.03	0.00	0.01	0.12	north_america
Zoop	0.03	0.02	0.00	0.00	0.05	north_america
ZooCube	0.05	0.01	0.00	0.00	0.07	north_america

Analysis:

We observe that some games perform best in specific regions.
For example:

- Zumba Fitness stands out in North America, with total global sales of 9.79 million, making it the most popular in this region.
- Zumba Fitness 2 also performs best in North America, reaching 2.88 million in global sales.
- Games like Zyuden Sentai Kyoryuger: Game de Gaburincho!! and Zwei!! perform best in Japan, with 0.05 and 0.02 million in sales, although these figures are quite low compared to the top-selling games in other regions.
- On the other hand, Zumba Fitness Rush and Zumba Fitness Core have a stronger presence in Europe, though their global sales are more limited.
- This pattern reflects the differences in tastes and preferences across regions, with North America dominating global sales for certain Zumba Fitness games, while Japan continues to show a strong preference for local games.

**Regions
Where
Games
Performed
Best**

Success of Platforms in Different Regions: A Comparative Analysis

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platform	total_na	total_eu	total_jp	total_others	biggest_sales
Wii	497.37	264.35	68.28	79.20	north_america
NES	125.94	21.15	98.65	5.31	north_america
GB	113.64	47.51	85.12	8.16	north_america
DS	388.56	194.07	175.02	60.29	north_america
X360	594.33	278.01	12.30	84.67	north_america
PS3	388.9	340.47	79.21	140.81	north_america
PS2	572.92	332.63	137.54	190.47	north_america
SNES	61.23	19.04	116.55	3.22	japan
GBA	184.12	74.59	46.56	7.61	north_america
3DS	78.03	58.30	97.30	12.55	japan
PS4	96.8	123.70	14.30	43.36	europe
N64	138.91	41.03	33.76	4.31	north_america
PS	334.71	212.38	139.78	40.69	north_america
XB	182.06	59.65	1.38	8.48	north_america
PC	92.1	137.64	0.17	24.33	europe

Analysis:

In this analysis, we observe how each platform performed best in different regions of the world:

- **North America** was the region where several platforms dominated, with Wii leading with 497.37 million in sales, followed by X360 (594.33 million) and PS2 (572.92 million). NES and GB also performed best in this region, with sales of 125.94 million and 113.64 million, respectively.
- **Japan**, which has a preference for classic platforms in the region, such as SNES (116.55 million) and 3DS (97.30 million).
- **Europe:** Showed a preference for the PS4, which had sales of 123.70 million, making it the second most popular platform in this region.
- Additionally, PC (137.64 million) had a stronger presence in Europe compared to other regions.

Region with the Most Sales Each Year: Analyzing Trends Over Time

Analysis:

Over the years, **North America** has been the dominant region in video game sales, standing out in almost every year from 1980 to 2015.

On the other hand, **Japan** led in certain years, such as in 1983, 1986, and 1991. In 2016, **Europe** surpassed the other regions for the first time, marking a shift in market trends.

Despite some ups and downs, North America has maintained its dominance over time.

# year	total_na	total_eu	total_jp	total_others	biggest_sale
1980	10.59	0.67	0.00	0.12	north_america
1981	33.4	1.96	0.00	0.32	north_america
1982	26.92	1.65	0.00	0.31	north_america
1983	7.76	0.80	8.10	0.14	japan
1984	33.28	2.10	14.27	0.70	north_america
1985	33.73	4.74	14.56	0.92	north_america
1986	12.5	2.84	19.81	1.93	japan
1987	8.46	1.41	11.63	0.20	japan
1988	23.87	6.59	15.76	0.99	north_america
1989	45.15	8.44	18.36	1.50	north_america
1990	25.46	7.63	14.88	1.40	north_america
1991	12.76	3.95	14.78	0.74	japan
1992	33.87	11.71	28.91	1.65	north_america
1993	15.12	4.65	25.33	0.89	japan
1994	28.15	14.88	33.99	2.20	japan
1995	24.82	14.90	45.75	2.64	japan
1996	86.76	47.26	57.44	7.69	north_america
1997	94.75	48.32	48.87	9.13	north_america
1998	128.36	66.90	50.04	11.03	north_america
1999	126.06	62.67	52.34	10.05	north_america
2000	94.49	52.75	42.77	11.62	north_america
2001	173.98	94.89	39.86	22.76	north_america
2002	216.19	109.74	41.76	27.28	north_america
2003	193.59	103.81	34.20	26.01	north_america
2004	222.59	107.32	41.65	47.29	north_america
2005	242.61	121.94	54.28	40.58	north_america
2006	263.12	129.24	73.73	54.43	north_america
2007	312.05	160.50	60.29	77.60	north_america
2008	351.44	184.40	60.26	82.39	north_america
2009	338.85	191.59	61.89	74.77	north_america
2010	304.24	176.73	59.49	59.90	north_america
2011	241.06	167.44	53.04	54.39	north_america
2012	154.96	118.78	51.74	37.82	north_america
2013	154.77	125.80	47.59	39.82	north_america
2014	131.97	125.65	39.46	40.02	north_america
2015	102.82	97.71	33.72	30.01	north_america
2016	22.66	26.76	13.70	7.75	europa
2017	0	0.00	0.05	0.00	japan
2020	0.27	0.00	0.00	0.02	north_america

name	na	eu	jp	others	global	mean	stddev_regions
Bayonetta 2	0.34	0.28	0.10	0.30	1.01	0.26	0.09
Persona 4: Golden	0.46	0.34	0.38	0.21	1.38	0.35	0.09
Resonance of Fate	0.32	0.35	0.30	0.12	1.07	0.27	0.09
Hot Shots Golf: Open Tee	0.5	0.50	0.63	0.33	1.96	0.49	0.11
Final Fantasy Tactics: The War of the Lions	0.45	0.25	0.30	0.16	1.16	0.29	0.11
Resident Evil: Revelations 2	0.31	0.43	0.32	0.13	1.18	0.3	0.11
Marvel vs. Capcom 2: New Age of Heroes	0.4	0.27	0.28	0.08	1.04	0.26	0.11
F-Zero: Maximum Velocity	0.39	0.16	0.37	0.12	1.04	0.26	0.12
Tenchu: Wrath of Heaven	0.44	0.34	0.25	0.11	1.14	0.28	0.12
Metal Gear Solid: Portable Ops	0.38	0.35	0.38	0.09	1.19	0.3	0.12

Games with the Most Balanced Sales by Region

Analysis

It is observed that games like Persona 4: Golden and Bayonetta 2 are examples of titles with relatively balanced success across various regions, as indicated by their low standard deviation (0.09 for both).

These games succeeded not only in one region but also in several, demonstrating global appeal and popularity across different regions. We also observe that Resident Evil: Revelations 2 and Marvel vs. Capcom 2: New Age of Heroes show balanced sales distributions, being popular in North America and Europe, as well as other regions of the world.

Games with the Least Balanced Sales by Region

Analysis:

We can see that games like Wii Sports and Super Mario Bros. have very high standard deviations (15.31 and 12.4, respectively). This means their sales are heavily concentrated in a single region, while in other regions, they show much lower figures in comparison. Duck Hunt and Tetris also reflect strong regional preferences, with significantly fewer sales outside their main markets. For instance, in North America, Duck Hunt and Tetris reached 26.93 and 26.17 million units sold, respectively, while in Europe, their sales dropped to only 0.63 million (Duck Hunt) and 2.95 million (Tetris).

name	na	eu	jp	others	global	mean	stddev_regiones
Wii Sports	41.49	29.02	3.77	8.46	82.74	20.69	15.31
Super Mario Bros.	32.48	4.88	6.96	0.99	45.31	11.33	12.4
Duck Hunt	26.93	0.63	0.28	0.47	28.31	7.08	11.46
Tetris	26.17	2.95	6.03	0.69	35.84	8.96	10.12
Grand Theft Auto V	23.46	23.04	1.39	8.03	55.92	13.98	9.56
Call of Duty: Black Ops	17.01	8.69	0.59	3.12	29.40	7.35	6.3
Call of Duty: Modern Warfare 3	15.58	11.29	0.62	3.35	30.83	7.71	6
Kinect Adventures!	14.97	4.94	0.24	1.67	21.82	5.46	5.75
Super Mario World	15.99	4.86	4.49	0.75	26.07	6.52	5.7
Call of Duty: Ghosts	14.92	9.05	0.48	2.92	27.38	6.84	5.61

Lowest Selling Games Worldwide

Analysis:

The lowest-selling games reached only 10,000 units sold. Most of these titles belong to lesser-known or niche platforms such as GBA, DS, PSP, and PSV.

name	global_sales	platform
Spirits & Spells	0.01	GBA
Dragon Zakura DS	0.01	DS
Hotel Giant DS	0.01	DS
Blackthorne	0.01	GBA
Chameleon: To Dye For!	0.01	DS
Super Battle For Money Sentouchuu: Kyuukyoku...	0.01	3DS
Turok	0.01	PC
DokuSui: DokiDoki Suikoden	0.01	PSP
Don Bradman Cricket 14	0.01	X360
Coven and Labyrinth of Refrain	0.01	PSV

Lowest Selling Genres Worldwide

Analysis:

The genre with the lowest total sales was Strategy, followed by Adventure and Puzzle. While these genres often have loyal fan communities, they don't generate as many sales as blockbuster genres like Action and Sports.

genre	total
Strategy	173.43
Adventure	234.80
Puzzle	242.22
Simulation	390.16
Fighting	444.05
Racing	726.77
Misc	797.62
Platform	829.15
Role-Playing	923.84
Shooter	1026.20
Sports	1309.24
Action	1722.88

Platforms with the Lowest Global Sales

Analysis:

The platforms with the lowest recorded sales were PCFX and GG, each with fewer than 50,000 units sold. These were followed by 3DO and TG16, both under 160,000 units. Some of these platforms targeted niche markets, while others simply failed to achieve commercial success.

platform	total
PCFX	0.03
GG	0.04
3DO	0.10
TG16	0.16
WS	1.42
NG	1.44
SCD	1.87
DC	15.97
GEN	28.36
SAT	33.59

Publishers with the Lowest Global Sales

publisher	total
EON Digital Entertainment	0.01
Interworks Unlimited, Inc.	0.01
Naxat Soft	0.01
Genterprise	0.01
Media Entertainment	0.01
UIG Entertainment	0.01
Boost On	0.01
Epic Games	0.01
New World Computing	0.01
Stainless Games	0.01

Analysis:

We can observe that some publishers recorded very low sales (less than 10,000 units), which may be due to several factors:

- Limited reach: A low capacity to connect with broader audiences.
- Few releases: Some of these publishers may have launched just a few games, and failed to achieve enough success to boost brand recognition.
- Niche market focus: Some companies target very specific audiences, which limits their potential for global sales.

Analysis of the Years with the Lowest Global Sales

year	total
2017	0.05
2020	0.29
1980	11.38
1983	16.79
1987	21.74
1982	28.86
1991	32.23
1981	35.77
1986	37.07
1993	45.98

Analysis:

We can see that the years with the lowest sales were during the early days of the gaming industry (from 1980 to 1993), when global sales did not even reach 50 million units. This was likely because gaming was a new industry that hadn't yet gained the popularity it would achieve later on.

We also notice that in more recent years (2017 and 2020), sales were extremely low (0.05 million in 2017 and 0.29 million in 2020), especially when compared to other years. As mentioned previously, this could be due to the rise of mobile gaming, which caused many consumers to shift from console gaming to playing on their smartphones.

Conclusions

- **North America's Dominance:** This region controls nearly 50% of the global market, proving itself as the most crucial player in the gaming industry.
- **Nintendo and EA Leadership:** These two publishers have been market leaders, consistently alternating at the top spot throughout most years.
- **2006–2011: The Golden Era of Gaming:** This period marked a peak in global video game sales, innovation, and franchise success.
- **High Sales Dispersion:** The market shows a wide spread, with significant outliers—some games selling extremely well while many others struggled.
- **The Power of SQL:** This project illustrates how SQL can be a powerful tool to uncover insights from real-world datasets across various industries.

Recommendations and Opportunities for Improvement

- **Project Strengths**

- **Key Market Insights:** The project successfully identified major trends and how video game sales are distributed across different regions of the world.
- **Efficient Use of SQL:** Advanced queries were implemented to extract statistics, rankings, and trends, showcasing SQL's potential for analyzing and managing large-scale datasets.

- **Areas for Improvement**

- **More Intuitive Visualizations:** Incorporating tools like Power BI or Tableau could enhance presentation with heat maps, line graphs, scatter plots, and other visual formats.
- **Deeper Statistical Analysis:** Applying advanced techniques such as ANOVA, regression, or correlation analysis could lead to more robust conclusions.

- **Project Expansion**

- **Updated Data:** Including more recent data could reflect the current state of the industry and increase the project's relevance.
- **Predictive Modeling:** Leveraging Python-based machine learning tools could help forecast future video game sales and market behaviors.

- **Practical Applications**

- **Strategic Decision-Making for Companies:** Video game companies could use this type of analysis to refine their strategies, target regions more effectively, and make data-driven business decisions.

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LinkedIn or GitHub!**

Data Source

The dataset used in this project was obtained from Kaggle, under the title "🎮 EDA - VIDEO GAME SALES".

This dataset compiles information from VGChartz.com, a well-known website for tracking and analyzing video game and console sales data.

 **Kaggle:** [EDA - VIDEO GAME SALES](#)

 **VGChartz:** vgchartz.com